

SIRATUL ISLAM

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Education

Shahjalal University of Science and Technology

Bachelor of Science (Hons) in Physics

Aug. 2023 – Aug 2027

Sylhet, Bangladesh

Experience

Linux Foundation - Zephyr RTOS Project

Remote

Biometrics, Authentication, and FIDO2 Subsystems Maintainer

Feb. 2026 – Present

- Author and Maintainer of Zephyr's biometrics subsystem (PR #100139), the first in an RTOS
- Author and Maintainer of Zephyr's FIDO2 authentication subsystem (PR #104327), the first in an RTOS
- Review community contributions to the Authentication / FIDO2 / Biometrics subsystems
- Technologies: Zephyr RTOS, Embedded C, Subsystem Architecture, Device Drivers, Device Tree, Git

Contributor (Triage)

Jun. 2025 – Present

- Zephyr Contributor with Triage access and 20+ merged PRs across ARM, RISC-V, and Xtensa platforms
- Develop device drivers and board support packages serving embedded developers globally
- Review community contributions and provide technical guidance on driver architecture and device tree bindings
- Maintain documentation for 9+ board support packages in Zephyr upstream

Linux Foundation - The Linux Kernel

Remote

Driver Maintainer, Industrial I/O (IIO) Subsystem

Mar. 2026 – Present

- **Maintained Drivers:** ST VL53L1X (Time-of-Flight sensor)
- **In-Review Drivers:** QST QMC5883L (Magnetometer)
- Authored the driver and Device Tree bindings and used latest Kernel APIs
- Review and manage version control via lore.kernel.org, and collaborate with contributors

Shahjalal University of Science and Technology

Jun 2025 – Present

Research Assistant - Department of Electrical and Electronic Engineering

Sylhet, Bangladesh

- Developing IoT-enabled smart control systems and radar-based occupancy detection for government energy optimization
- Building RFID-based attendance management systems for rural educational institutions in Bangladesh
- Building STM32F4 control system, implementing ROS2 nodes, developing OpenCV vision algorithms

Hackules Inc.

Jun. 2024 – Jun. 2025

Software Engineer

Remote

- Led full-stack/frontend development for educational platforms (Opedemy, Teachers Today) serving 1000+ active users

Selected Projects

Autonomous Vehicle Navigation System | ROS2, C++, Jetson Orin, STM32F4

ongoing research

- EEE department-funded project: ROS2-based autonomous navigation with LiDAR fusion and OpenCV computer vision
- Developed STM32F4 vehicle control system for real-time actuator management and motor control with sensor fusion

STM32 BME280 HAL Driver | C, STM32, I2C, Bare-Metal

source code

- Developed a custom Hardware Abstraction Layer (HAL) driver in C for the Bosch BME280 environmental sensor
- Implemented reliable I2C communication protocols to parse temperature, humidity, and pressure telemetry

TM1637 Linux Kernel Module | C, Makefile, Device Tree

source code

- Developed an out-of-tree (OOT) Linux loadable kernel driver for the TM1637 7-segment display
- Implemented sysfs interfaces and hardware communication layers for user-space interaction and brightness control

Technical Skills

Specializations : Zephyr RTOS, Linux Kernel, FIDO2/CTAP2, Embedded Biometrics, IoT Security & Hardware Auth

Embedded Systems: Zephyr RTOS, FreeRTOS, Linux Kernel, STM32 HAL, ESP-IDF, Embedded Linux, ROS2

Platforms & Protocols : STM32, ESP32, nRF52, Raspberry Pi, MQTT, BLE, Wi-Fi, LoRa, USB, NFC, I2C, SPI, UART

Development Tools: Embedded C/C++, Rust, STM32CubeIDE, CMake, PlatformIO, OpenOCD, KiCAD, Git

Full-Stack Web: C#, TypeScript, .NET, SvelteKit, Next.js, PostgreSQL, Docker, Podman

Publications

1. S. Islam, M. Rasedujjaman, "BASE: A Multi-Modal Biometric Architecture for Resource-Constrained RTOS," *Elsevier Internet of Things*, under review, 2026.